

A Major Project Presentation on AI Powered Job Portal



Presented By:

Dheeraj Sengar

0928CS221056

Btech (CSE) , 6th Sem

Guided By:

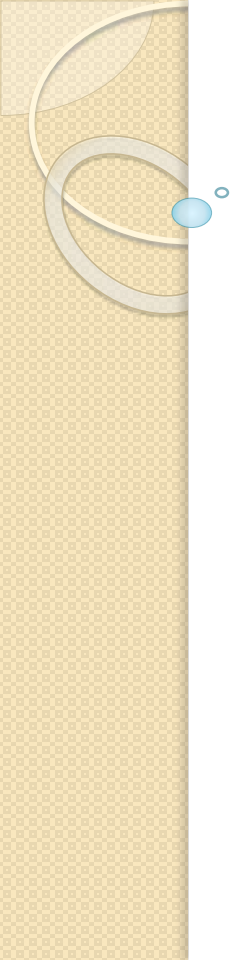
Mr . Raju Sharma

Head, CSE

IPS-CTM ,Gwalior

Contents:

- Introduction To AI Job Portal
- Features of AI Job Portal
- Use Case Diagram
- Data Flow Diagram
- Tech Stack
- Project Screenshots
- Acknowledgement



Introduction To AI Job Portal

The AI-Powered Job Portal is a **next-generation recruitment platform** inspired by Naukri and LinkedIn.

It leverages **AI and ML** to improve the hiring process for both **candidates** and **recruiters**.

Provides **smart job recommendations**, **salary prediction**, and **resume-based candidate ranking**.

Built using **MERN stack** for scalability and **Python ML models** for intelligence.

Features of AI Job Portal

Smart Job Recommendations

- **AI Part:** Collaborative filtering + NLP-based skill matching.
Recommend jobs based on skills, experience, and browsing history.

Workflow:

Candidate uploads profile → AI extracts skills → Suggests matching jobs.

Salary Prediction for Role & Location

AI Part: Regression models (Linear Regression, Random Forest, XGBoost).

Dataset: role, experience, location, company → predict expected salary.

Workflow:

User searches “Software Engineer, Bangalore” → AI predicts salary range.

Resume-based Candidate Ranking (for Recruiters)

AI Part: NLP (TF-IDF, BERT, or embeddings).

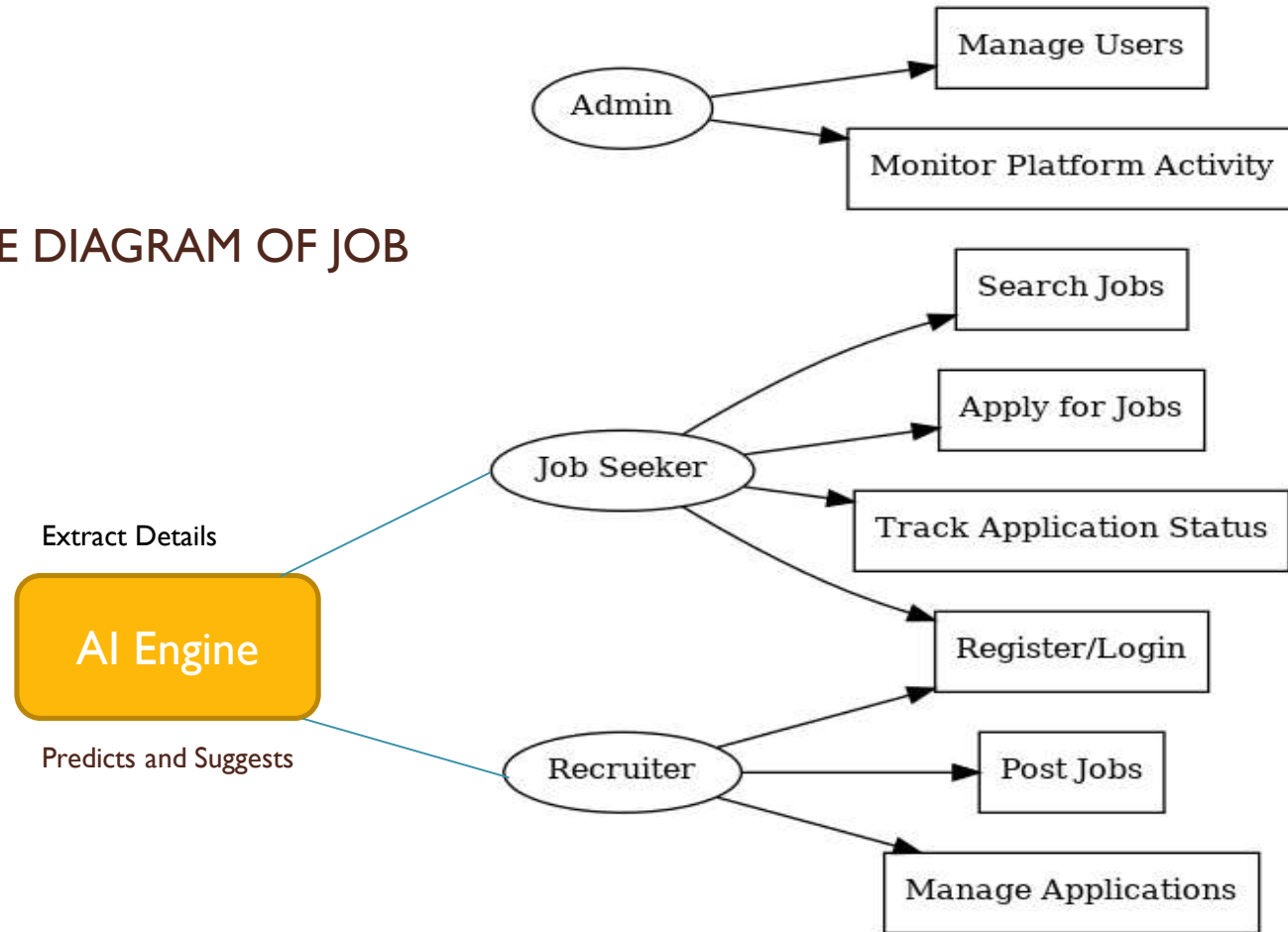
Parse resumes → extract keywords (skills, roles, education).

Rank candidates vs. job description.

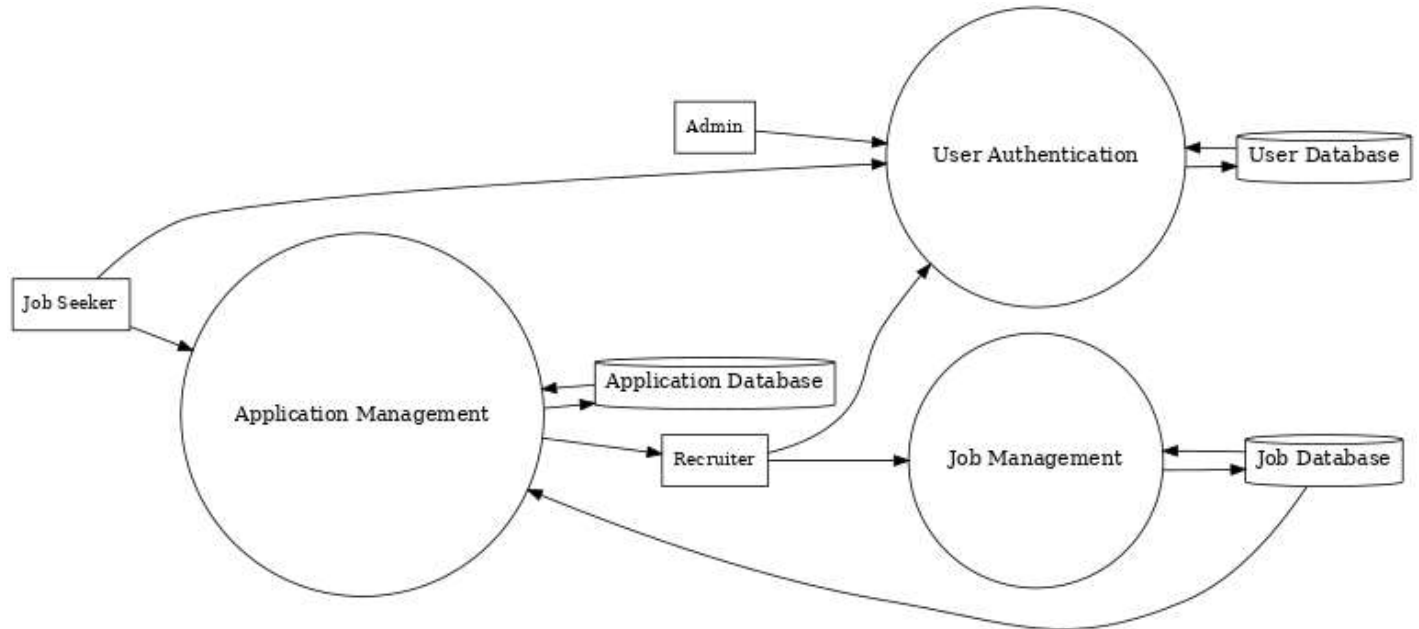
Workflow:

Recruiter posts job → Uploads resumes → AI ranks candidates by suitability.

USE CASE DIAGRAM OF JOB PORTAL

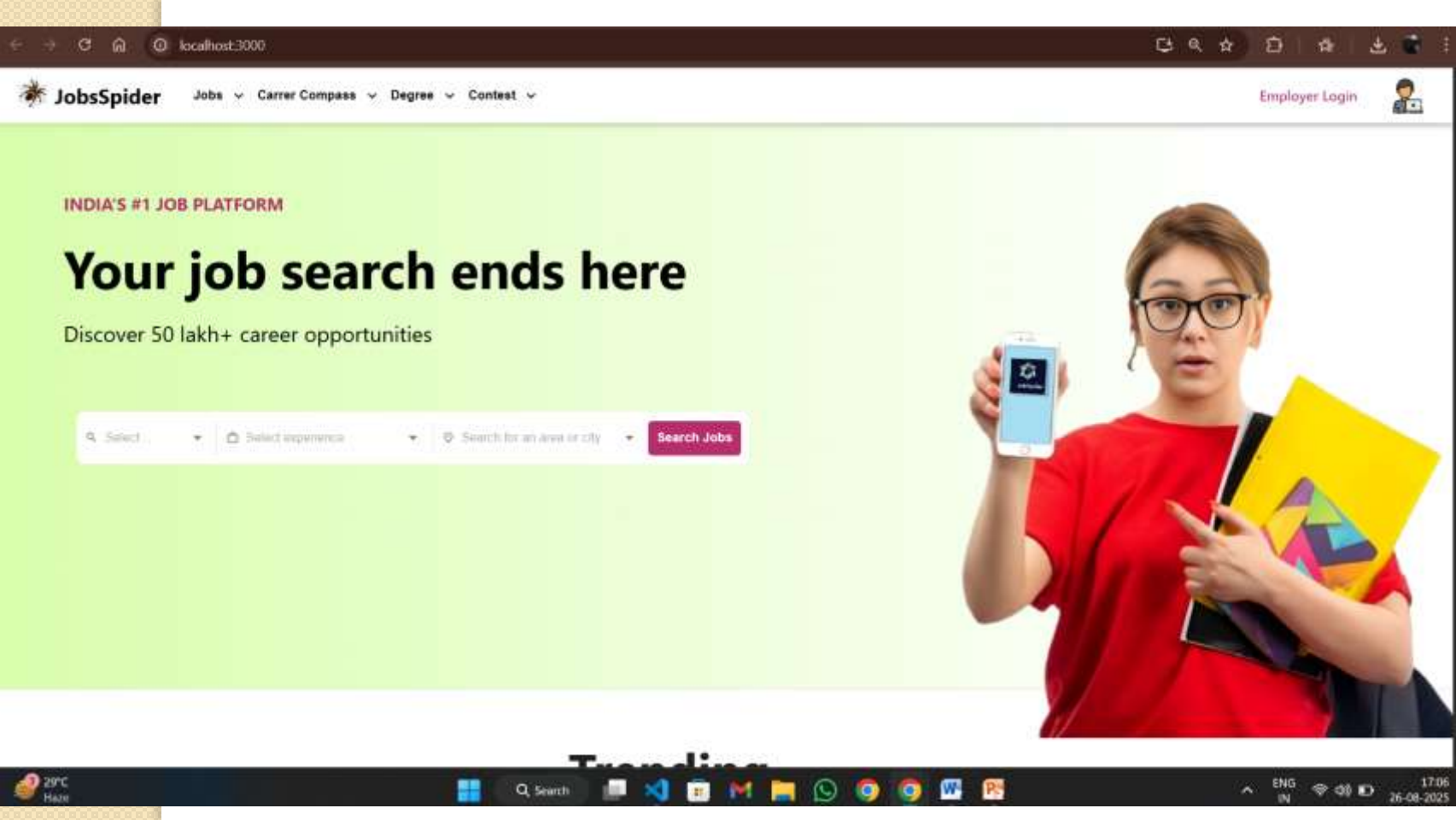


DATA FLOW DIAGRAM



Tech Stack

- ◆ **Frontend:** React + CSS
- ◆ **Backend:** Node.js (Express)
- ◆ **Database:** SQL (MySQL / PostgreSQL) for Users, Jobs, Applications, Resumes
- ◆ **AI/ML Models:** Python (scikit-learn, TensorFlow, HuggingFace for NLP)
- ◆ **Integration:** Flask / FastAPI microservice for AI, connected with Node.js backend
- ◆ **Deployment:** Docker + AWS/GCP





🔍 MERN ▾

🏠 Fresher ▾

📍 Hyderabad ▾

Search Jobs

Showing 210 jobs based on your filter

Filters

Clear all

Experience

Your work experience

Freshers



0 years

11 years

Date Posted

Your work experience

☐ All

☐ Last 24 hours

☐ Last 3 Days

☐ Last 7 Days

Salary

Salary you expect

Rs



Urgent Hiring



Blinkit
Comm
Private
Limited

IT Full Stack

Blinkit Commerce Private Limited



Hyderabad, Vijayawada, Bangalore

₹ 65000 to ₹ 110000

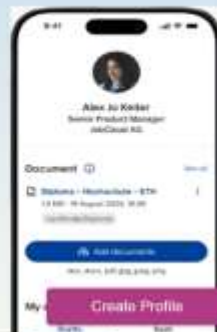
Fulltime

3AM-6PM

Prev 1 2 3 4 5 10 Next

Login with Apna and experience more!

- Personalised job matches
- Direct connect with HRs
- Latest updates on the job



/* LOGO */

Sign in

Email

your@email.com

Password

[Forgot your password?](#)

☐ Remember me

SIGN IN

Don't have an account? [Sign up](#)

or



Company Register



H1 H2 Sans Serif : Normal : B I U S " | | | | | | | |

About Company



UPLOAD COMPANY LOGO

SAVE

RESET

ChatGPT

New Tab

React App

localhost:3001/companyjobs

 **JobsSpider**
Company jobs registration

Company ID

Category ID

SubCategory Id

Skills

Education Qualification

experience

H1 H2 Bold Italic Normal Bulleted Numbered Indent Outdent Link

JobDetail

jobType

MinSalary

MaxSalary

Schedule

WorkLocation City

H1 H2 Bold Italic Normal Bulleted Numbered Indent Outdent Link

Benefit

H1 H2 Bold Italic Normal Bulleted Numbered Indent Outdent Link

SupplementalPay

join date

Application Deadline

Expected Start

dd-mm-yyyy

dd-mm-yyyy

dd-mm-yyyy

H1 H2 Bold Italic Normal Bulleted Numbered Indent Outdent Link

Application Question

Contact Person

Email Address

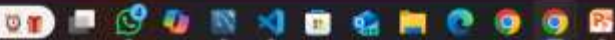
Mobile No

SAVE

RESET

20°C
Hazy

Search



ENG
IN

29-11-2024

10:33

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- database
- flight_management
- food_management
- jobspider
 - Tables
 - category
 - companies
 - requiredskills
 - state
 - subcategory
 - city
- sys

Administration Schemas Information

Table: city

Columns:

- cityid: int PK
- stateid: int
- cityname: varchar(45)

Object Info Session

state city state city subcategory state companies city companies requiredskills

Limit to 1000 rows

1 SELECT * FROM jobspider.city;

Result Grid

cityid	stateid	cityname
300	1	Gwalior
301	1	Indore
302	1	Bhopal
303	2	Lucknow
304	2	Kanpur
305	2	Prayagraj
306	2	Varanasi

city | x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:06:43	SELECT * FROM jobspider.requiredskills LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
2	11:03:48	SELECT * FROM jobspider.city LIMIT 0, 1000	6 row(s) returned	0.016 sec / 0.000 sec



| Contact and Feedback

Thank You Page

We value your feedback and are here to assist you with any inquiries. Connect with us today